Question Bank for DISL

1	"Managing customer orders" system. Following is the Scenario: 1. a customer has a unique customer number and contact information , 2. a customer can place many orders, but a given purchase order is placed by one customer 3. a purchase order has a many-to-many relationship with a stock item. Write a database trigger to update a stock table when a record is inserted in contains table.
2	Consider Customer database for collection 1. Use index on collection 2. Calculate amount per customer using aggregation /Map reduce operation using MongoDB.
3	Create Teacher Database in Mongodb Database. Which contain the information of Teacher_id, name of a teacher, department of a teacher, salary and status of a teacher? Here status is whether teacher is approved by the university or not. • Implement all the basic queries on the Teacher Database using mongodb commands. • Find out the teacher who are having salary greater than 25000 • Modify the E &TC department teacher salary with 10% growth • Display the teacher information according to status • Implement map reduce function to display average salary of teacher in each department.
4	Consider the relational database : dept (dept-no, dname, LOC) emp (emp-no, ename, designation,sal) project (proj-no, proj-name, status) dept and emp are related as 1 to many. project and emp are related as 1 to many. Write sql expressions for the following : i) List all employees of 'INVENTORY' department of 'PUNE' location. ii) Give the names of employees who are working on 'Blood Bank' project. iii) Give the name of managers from 'MARKETING' department. iv) Give all the employees working under status 'INCOMPLETE' projects

5	Create a table 'emp' with the following columns by assuming suitable data type and size with correct syntax in SQL. Emp–id, Ename, City, State, Salary, Age, Hire_ date. Give an expression in SQL to solve each of the following queries: i) Find the names of all employees whose name starts with 'Sa'. ii) List all the employees name and salary whose age is less than 40 years. iii) Select the employees whose salary is between Rs. 20000 and Rs. 30000. iv) Write a procedure to display highest salary of an employee without using aggregate functions
6	Create a collection 'emp' with following fields Emp–id, Ename, City, State, Salary, Age, Hire_ date. Give an expression in NoSql to solve each of the following queries: i) Find the names of all employees whose name starts with 'Sa'. ii) List all the employees name and salary whose age is less than 40 years. iii) Select the employees whose salary is between Rs. 20000 and Rs. 30000. iv) Write a function to display highest salary of an employee.
7	A database consists of following tables. PROJECT(PNO, PNAME, CHIEF) EMPLOYEE(EMPNO, EMPNAME) ASSIGNED(PNO,EMPNO) A. Get count of employees working on project. B. Get details of employee working on project pr002. C. Get details of employee working on project DBMS. D. Write a trigger to delete all corresponding records from assigned table if employee id deleted.

	E. Write a trigger to keep back up of assign table records if project is deleted.
8	Create a database for Employee = (emp_no, emp_name, hiredate, comm., netsal ,dept_no , designation) 1. Display all the employee details in department 30. 2. List the names, numbers and departments of all clerks. 3. Find the employees whose commission is greater than their salaries. 4. Find the employees whose commission is greater than 60% of their salaries. 5. List the name job and salary of all the employees in department 20 who earn more than 2000/-. 6. Find all the clerks in department 30 whose salary is greater than 1500/-. 7. Find all employees whose designation is either manager or president. 8. Find all managers who are not in department 30. 9. Find all the details of all the managers and clerks in department 10. 10. Find the details of the managers in department 10 and all clerks in department 20.
9	Consider the relational database Supplier (sid, sname, address) Parts (pid, pname, color) Catalog (sid, pid, cost) Write SQL queries for the following: i) Find names of suppliers who supply some red parts. ii) Find names of all parts whose cost is more than Rs. 25 iii) Find name of all parts whose color is green. iv) Find name of supplier and parts with its color and cost. Implement any one pl/sql statement for the same statement.

10	You need to create a movie database. Create three tables, one for actors(AID, name), movies(MID, title) and actor_role(MID, AID, rolename). Use appropriate data types for each of the attributes, and add appropriate primary/foreign key constraints. 1. Insert data to the above tables (approx 3 to 6 rows in each table), including data for actor "Charlie Chaplin", and for yourself (using your roll number as ID). 2. Write a query to list all movies in which actor "Charlie Chaplin" has acted, along with the number of roles he had in that movie. 3. Write a query to list all actors who have not acted in any movie 4. List names of actors, along with titles of movies they have acted in. If they have not acted in any movie, show the movie title as null.
11	Employee (person-name, street, city) works (person-name, company-name, salary) Company (company-name, city) Manages (person-name, manager-name) Consider the above relational database. Write SQL queries for the following: 1. Find the names, street address, and cities of residence of all employees who work for First Bank Corporation and earn more than $10,000 per annum. 2. Find the names of all employees in this database who live in the same city as the company for which they work. 3. Find the names of all employees who live in the same city and on the same street as do their managers. 4. Write a Trigger on update of employee company_name
12	Consider following database: Student (Roll_no, Name, Address)

	Subject (Sub_code, Sub_name) Marks (Roll_no, Sub_code, marks) Write following queries in SQL: • Find average marks of each student, along with the name of student. • Find how many students have failed in the subject "DBMS". • Write a Trigger that check the rollno must be start with 'TE'.
13	Book = { Book_No ,Book_Name, Author_name , Cost, Category} Member = { M_Id , M_Name ,Mship_type, Fees_paid,Max_Books_Allowed, Penalty_Amount } Issue ={Lib_Issue_Id , Book_No , M_Id, Issue_Date, Return_date} • List top 5 books which are issued by Annual members • List the names of members who has issued the books whose cost is more than 300 rupees and whose author is "Scott Urman" • Write a query to display number of booked in each category of books issued by all member types.
14	Design and Develop DB for "Customer_ order" with all constraints(Not NULL, PrimaryKey, Foreign Key). Customer (Cust_no, name, Street, city, state)-Primary key (Cust_no) Order(Order_no,Cust_no,Order_date,Ship_date,Tocity,ToState,ToZip)-Primarykey(order_no),Foreignkey(Cust_no) Containd(Order_no,Stock_no,quantity,Discount)-Foreignkey(order_no),(Stock_no) Stock(Stock_no,price,tax)-Primarykey(stock_no) • Create index on table order with column order_no. • Perform Alter Table Operation • Create View on columns Order_no and Customer_no with order table.
•	• Design the DB for "Pets" in Mongodb and perform following operations: Pet(pet_name, owner,sex, birth_date, death_date) • Insert, Update, Delete

•	• Display all the pets information. • Display all pets in ascending order according to birth date. • Display all pets in descending order according birth date. • Display the pet information of specific owner. • Validate the data wherever require
16	Create a collection Tutorial(tutorial_id,tut_title,author,submission_date) Implement following operation on above collection 1) Insert 10 tutorial in collection 2) Update the date of specific tutorial 3) Delete tutorial from collection 4) Display all the tutorial details 5) Find specific tutorial_id and author of respective tutorial. 6) Display all the records of tutorial where author name starts with “a” 7) Display all tutorials in Ascending and Descending order according to tutorial_id or author name. 8) validate the data before entering into collection
17	Design & Develop DB for “Order Management System” with all the constraints. (At least 3 entities and relationships between them.) • Display all the Purchase orders of a specific Customer. • Get Customer and Data Item Information for a Specific Purchase Order. • Get the Total Value of Purchase Orders. • List the Purchase Orders in descending order as per total. • Display the name of customers whose first name starts with “A”. (String matching :Like operator) • Get the total no of customers. • Display average purchase amount of all the customers. • Display total purchase amount of all the customers.
18	Design & Develop DB for “Order Management System” with all the constraints. (At least 3 entities and relationships between them.)

	Execute queries like • Display the name of customer whose order amount is greater than all the customers. • Display order details of customer whose city name is “Pune” and purchase date is “22/08/2016” • Add discount of 5% to all the customers whose order is more than Rs. 10000/-. • Delete Purchase Order 1001. • Find DAYNAME, MONTHNAME and YEAR of the purchase order made on “1-11-2017” • Get current date & time, current time, current date • Find purchase details of the customers group by product category. • Find the purchase details of all the customers who made shopping today.
19	Create a NOSQL DB on “Movie”(Movie_id, name, type, budget ,date_of_release) using MongoDB and implement following operations on document. • Insert (batch insert, insert validation) the single and multiple documents • Remove the documents • Update the type of movie • Upserts • Use aggregate function to retrieve the big budget, small budget movie name. • Find the average budget of year “2017”
21	Create a NOSQL DB on “student”(roll_no, name, marks, residential_address,college_address) using MongoDB .Execute at least 10 queries on above MongoDB database that demonstrates following: • Find the student who has live in same city as college. • Retrieve the 4, 5 documents from collection. • Update the marks array for SE, TE, BE marks. • Use index, explain it and drop the index • Find the average of each year for every student.
22	Implement the aggregation and indexing with suitable example on above MongoDB database. • Aggregation • Create and drop different types of indexes and explain () to show the advantage of the indexes.

23	Consider the following database: Employees (Employee_id,first_name , last_name , email, ph_no , hire_date, job_id, Salary, department_id) Works(Employee_id,manager_id) Departments (Department_id,dept_name , location_id) Jobs (Job_id, job_title ,min_salary , max_salary) Locations (Location_id , street, city, state , country) Job_history(Employee_id , hire_date, leaving_date, salary, job_id, department_id) 1] Display all the employees in descending order of their salary. 2] Display employee_id, full name and salary of all employees who have joined in year 2006 according to their seniority. 3] List name of all departments in location 20,30 and 50 4] Display the full name of all employees whose first_name or last_name contains ‘a’ 5] Find the department with the most employees. 6] Write a procedure that accepts deptno value from a user, selects the maximum salary and minimum salary paid in the department, from the EMP table
24	Consider following database: Student (Roll_no, Name, Address) Subject (Sub_code, Sub_name) Marks (Roll_no, Sub_code, marks) Write following queries in MongoDB: Consider wherever embedded collection is required • Find average marks of each student, along with the name of student. • Find how many students have failed in the subject “DBMS”. • Find the students who get marks greater than 75 and and also find student who get

	less than 40. • Find the student whose addresses are ‘PUNE’.
25	Convert the following ER diagrams into Relational Models. Write DDL,DML,DQL statement for all and insert data into it.